

LOSO Experiment B_ypr_accel

YPR + Accel (both hands), no Nothing

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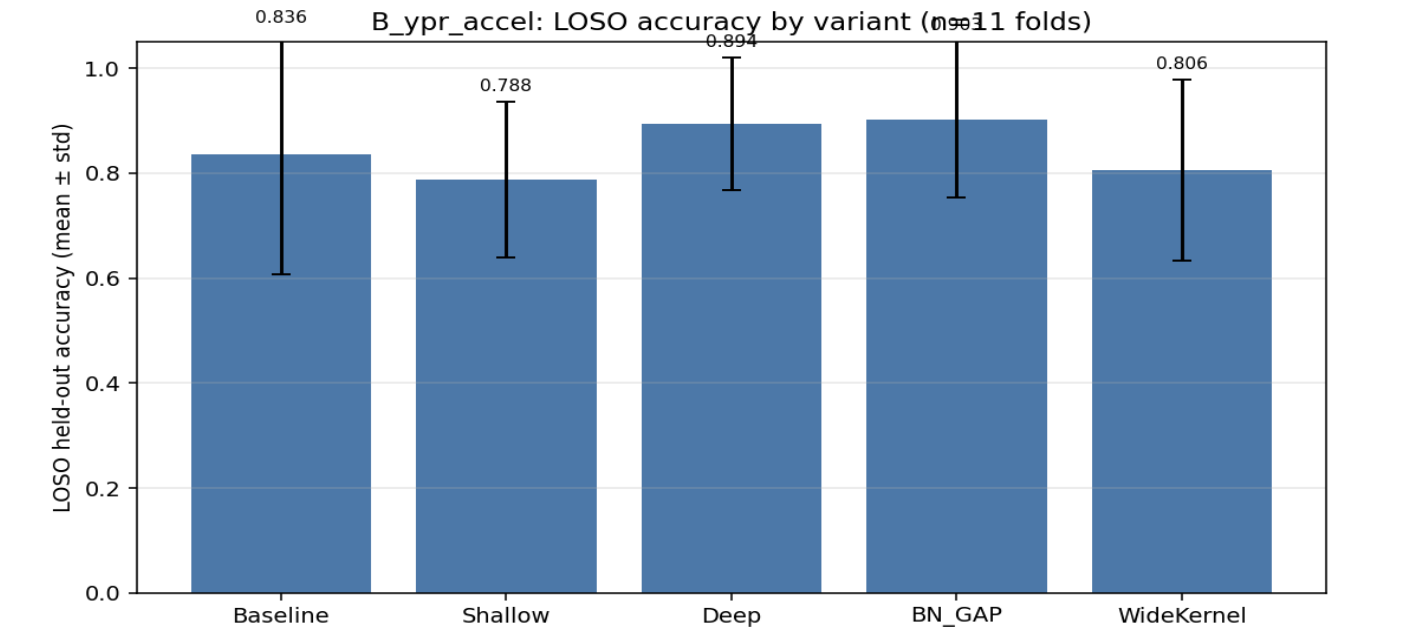
Experiment configuration

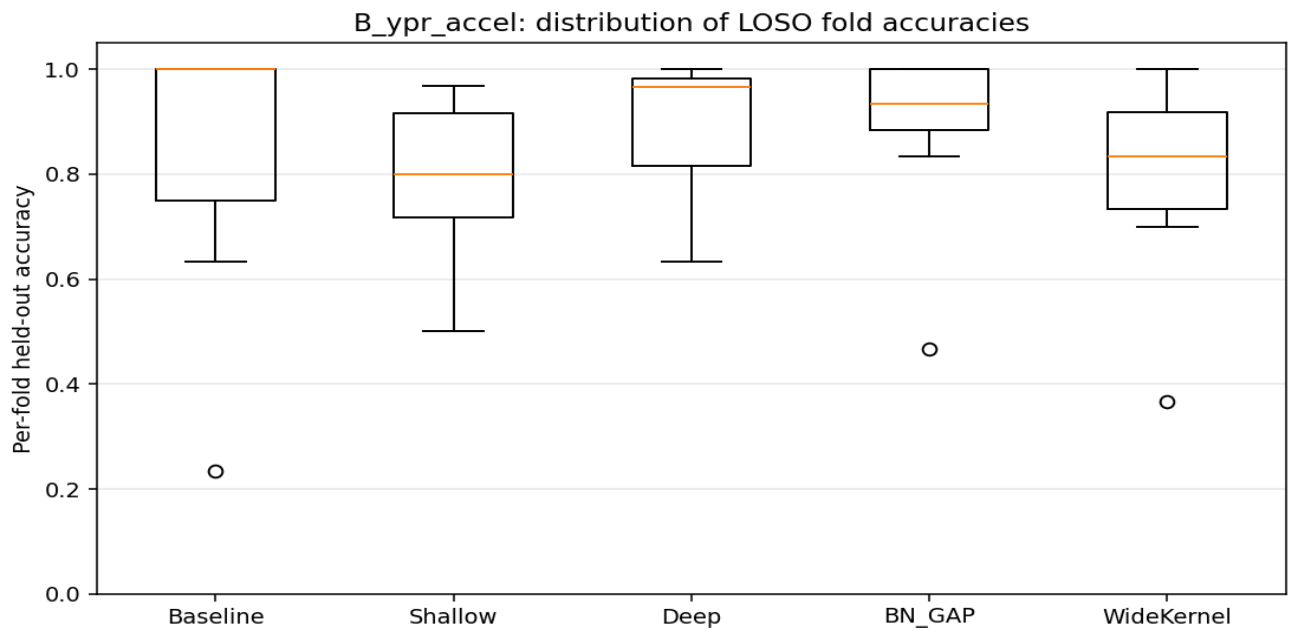
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	YPR, accel
Hands	left, right
Sensor channels	156
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	11: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

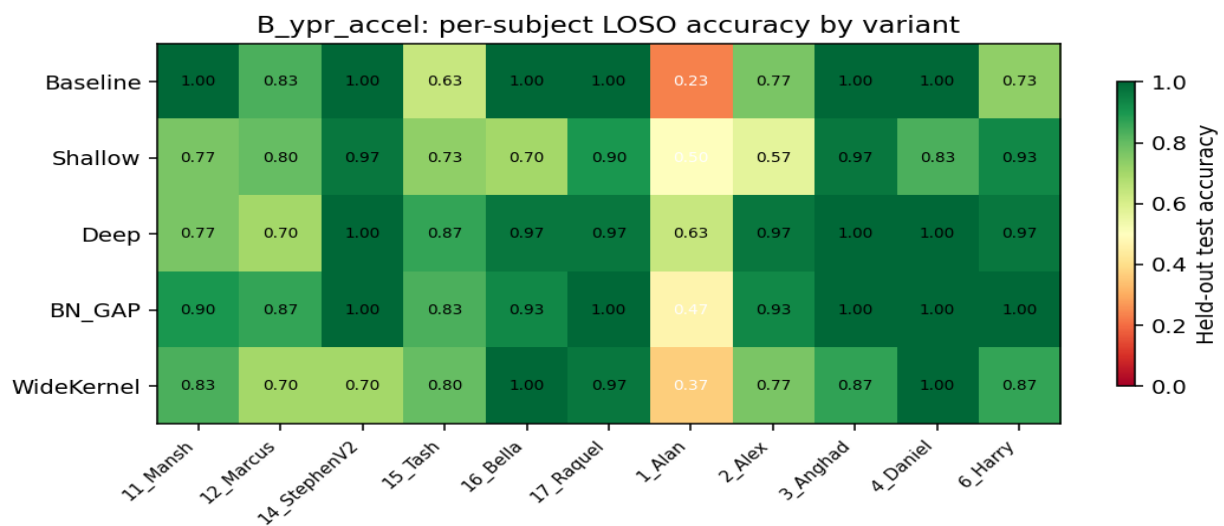
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
Baseline	110,630	0.8364	0.2298	0.2333	1.0000	0.9686
Shallow	69,254	0.7880	0.1487	0.5000	0.9677	0.8846
Deep	201,318	0.8939	0.1270	0.6333	1.0000	0.9720
BN_GAP	54,630	0.9030	0.1494	0.4667	1.0000	0.9898
WideKernel	126,502	0.8065	0.1724	0.3667	1.0000	0.9118

Best variant by LOSO mean: **BN_GAP**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	Baseline	Shallow	Deep	BN_GAP	WideKernel
11_Mansh	301	1.000	0.767	0.767	0.900	0.833
12_Marcus	301	0.833	0.800	0.700	0.867	0.700
14_StephenV2	301	1.000	0.967	1.000	1.000	0.700
15_Tash	301	0.633	0.733	0.867	0.833	0.800
16_Bella	301	1.000	0.700	0.967	0.933	1.000
17_Raquel	301	1.000	0.900	0.967	1.000	0.967
1_Alan	301	0.233	0.500	0.633	0.467	0.367
2_Alex	301	0.767	0.567	0.967	0.933	0.767
3_Anghad	300	1.000	0.968	1.000	1.000	0.871
4_Daniel	301	1.000	0.833	1.000	1.000	1.000
6_Harry	301	0.733	0.933	0.967	1.000	0.867

Variant: Baseline

Params: 110,630

LOSO mean accuracy: 0.8364 ± 0.2298 (min 0.2333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	5/5	5/5	5/5	1.000
12_Marcus	5/5	5/5	5/5	5/5	5/5	0/5	0.833
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
15_Tash	4/5	5/5	5/5	5/5	0/5	0/5	0.633
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	0/5	0/5	0/5	2/5	0/5	0.233
2_Alex	5/5	4/5	5/5	5/5	3/5	1/5	0.767
3_Anghad	5/5	5/5	5/5	5/5	6/6	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	2/5	5/5	5/5	0/5	5/5	0.733
MEAN	0.982	0.836	0.909	0.909	0.727	0.655	0.836

Variant: Shallow

Params: 69,254

LOSO mean accuracy: 0.7880 ± 0.1487 (min 0.5000, max 0.9677)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	5/5	5/5	0/5	3/5	5/5	0.767
12_Marcus	4/5	5/5	5/5	5/5	5/5	0/5	0.800
14_StephenV2	4/5	5/5	5/5	5/5	5/5	5/5	0.967
15_Tash	5/5	5/5	5/5	5/5	2/5	0/5	0.733
16_Bella	5/5	5/5	1/5	0/5	5/5	5/5	0.700
17_Raquel	5/5	5/5	5/5	5/5	5/5	2/5	0.900
1_Alan	5/5	4/5	1/5	0/5	5/5	0/5	0.500
2_Alex	0/5	4/5	5/5	0/5	5/5	3/5	0.567
3_Anghad	5/5	5/5	5/5	4/5	6/6	5/5	0.968
4_Daniel	5/5	5/5	5/5	0/5	5/5	5/5	0.833
6_Harry	5/5	5/5	5/5	4/5	4/5	5/5	0.933
MEAN	0.873	0.964	0.855	0.509	0.891	0.636	0.788

Variant: Deep

Params: 201,318

LOSO mean accuracy: 0.8939 ± 0.1270 (min 0.6333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	1/5	5/5	5/5	2/5	5/5	0.767
12_Marcus	5/5	1/5	5/5	5/5	5/5	0/5	0.700
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
15_Tash	4/5	5/5	5/5	5/5	2/5	5/5	0.867
16_Bella	5/5	5/5	5/5	4/5	5/5	5/5	0.967
17_Raquel	4/5	5/5	5/5	5/5	5/5	5/5	0.967
1_Alan	4/5	5/5	0/5	5/5	5/5	0/5	0.633
2_Alex	5/5	4/5	5/5	5/5	5/5	5/5	0.967
3_Anghad	5/5	5/5	5/5	5/5	6/6	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	4/5	5/5	0.967
MEAN	0.945	0.836	0.909	0.982	0.873	0.818	0.894

Variant: BN_GAP

Params: 54,630

LOSO mean accuracy: 0.9030 ± 0.1494 (min 0.4667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	2/5	5/5	5/5	5/5	5/5	0.900
12_Marcus	5/5	5/5	5/5	5/5	5/5	1/5	0.867
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
15_Tash	4/5	5/5	5/5	5/5	5/5	1/5	0.833
16_Bella	5/5	5/5	3/5	5/5	5/5	5/5	0.933
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	3/5	0/5	1/5	0/5	0.467
2_Alex	5/5	4/5	4/5	5/5	5/5	5/5	0.933
3_Anghad	5/5	5/5	5/5	5/5	6/6	5/5	1.000
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
MEAN	0.982	0.927	0.909	0.909	0.927	0.764	0.903

Variant: WideKernel

Params: 126,502

LOSO mean accuracy: 0.8065 ± 0.1724 (min 0.3667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	4/5	3/5	5/5	4/5	5/5	4/5	0.833
12_Marcus	5/5	2/5	5/5	1/5	5/5	3/5	0.700
14_StephenV2	5/5	5/5	5/5	0/5	3/5	3/5	0.700
15_Tash	5/5	5/5	5/5	5/5	3/5	1/5	0.800
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	4/5	5/5	5/5	5/5	5/5	5/5	0.967
1_Alan	5/5	0/5	0/5	1/5	5/5	0/5	0.367
2_Alex	5/5	4/5	5/5	0/5	5/5	4/5	0.767
3_Anghad	5/5	5/5	5/5	5/5	6/6	1/5	0.871
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	2/5	5/5	5/5	4/5	5/5	0.867
MEAN	0.964	0.745	0.909	0.655	0.909	0.655	0.806